

Additive Baselines on the 025 Build

The trigenic ladder under a random and a query-pair-disjoint split, in preparation for the main text

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Abstract

Six baselines that cannot represent a three-way interaction, or can represent only a per-screen offset, are fit on exactly the 376,732 trigenic records and the two splits the 025 transformer arms train on. Under the published random-over-records split the per-gene additive ridge reaches 0.400 test Pearson, a model that sees only which Kuzmin query double a triple carries reaches 0.390, and the transformer checkpoints reach 0.438 to 0.455. Holding out whole query doubles collapses the additive ridge to 0.185 on the one disjoint split the transformer trains on, and to 0.127 ± 0.033 over five disjoint folds, and drops the nonlinear baseline below the additive one. Four transformer runs have trained on the disjoint split, one in the 010 configuration and three with a fixed sequence embedding and a constant learning rate. Every one of them peaks on validation between epoch 2 and epoch 7, at 0.199 to 0.270 against an additive validation null of 0.150, and every one falls back afterward; the 010 configuration ends below the null at 0.131. None has a test score, so whether the transformer’s margin over additivity survives held-out screens is not yet measured. This is a working document: the figures fix the layout the main-text figure will have, and the panel and rows still to be filled are marked.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Provenance: ▲ not in the library mirror; verify at the linked source ● from a review’s summary table, not the primary paper

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1 The question, and the state of the evidence ✗

A model that claims to learn genetic interaction has to beat a null that cannot represent interaction, on data where the null cannot borrow the answer from the training set. The 010 trigenic model was never measured against such a null, and its published split lets every model score the identity of the screen a record came from. Both gaps are closed on the 025 build, which holds the same records as 010 with bit-identical labels, under two splits: the published random split (arm R) and a split that never shares a Kuzmin query double between parts (arm Q).

Measured. The six nulls on both arms, on test (Table 3, Figure 1a,b). Arm R reproduces the 010 numbers exactly. On arm Q the additive ridge falls from 0.400 to 0.185, the screen-only model to zero, and the nonlinear baseline from 0.432 to 0.141, below the ridge. Four transformer runs on arm Q, all with validation curves that peak by epoch 7 and decline (Figure 2a,b).

Not measured. Any transformer test score on arm Q. Replicates of the 010 configuration on arm Q. Figure 2c is the placeholder for both.

Vocabulary. A **query pair** is the double mutant a Kuzmin trigenic screen was built around; every triple in the build is one query pair plus one array gene, and 420 query pairs recur across the 376,732 records. A **screen** is the set of triples sharing a query pair. **Arm R** and **arm Q** are the two transformer training arms of experiment 025 and the two splits the baselines are fit under. The **ladder** is the ordered set of baselines B0 through B5 and the transformer, read left to right by interaction capacity. A **validation maximum** is the largest validation Pearson over a run’s logged epochs; it is an upward-biased order statistic and is never compared to a test number without saying so.

2 Data and the two splits ✗

The label. Write f_i , f_{ij} and f_{ijk} for the fitness of the single, double and triple mutant relative to wild type, and $\varepsilon_{ij} = f_{ij} - f_i f_j$ for the digenic interaction. The trigenic interaction score is

$$\tau_{ijk} = f_{ijk} - f_i f_j f_k - f_i \varepsilon_{jk} - f_j \varepsilon_{ik} - f_k \varepsilon_{ij}, \quad (1)$$

▲ [external: Kuzmin et al. 2018, 10.1126/science.aao1729; the definition of the released adjusted score, not re-derived here] a **third-order residual**: the multiplicative part of fitness and every pairwise term are subtracted before the number is reported, so τ is by construction the part of the triple phenotype that no single or double predicts. The loaders store it as **gene_interaction**.

The records. Subset S0 of the 025 build is its 376,732 trigenic records over 4,352 genes, the 010 build record for record: matching by genotype recovers every one, 376,526 labels are bit-identical and the remaining 206 differ by at most 5.6×10^{-17} (experiments/025-solid-growth/scripts/label_parity_010_vs_025.py). Every model below is fit on these records and nothing else.

The screen structure. Counting unordered gene pairs over the records finds 739,315 distinct pairs, of which 420 occur five or more times. Those 420 cover 376,733 pair-instances, one more than the record count, so every record contains exactly one recurring pair and a single record contains two. The 420 pairs are the Kuzmin query doubles. Grouping records by them recovers the screen exactly, and any model with a per-pair offset can absorb whatever is common to a screen.

The two arms. Arm R is 010’s own split, random over records, carried into the 025 index space by genotype identity. Arm Q assigns whole query pairs to parts, so no screen appears in more than one part. The two arms hold out nearly the same number of records and differ only in what is held out.

Table 1: The two arms. Both partition the same 376,732 records. Arm R’s parts each carry every one of the 420 query pairs; arm Q’s parts carry disjoint sets of them. The split artifacts are the files the trainer loads, under experiments/025-solid-growth/results/.

	Arm R	Arm Q
Held-out unit	record	query pair
Recurring query pairs	420, shared by every part	420, no pair in two parts
Query pairs, train / val / test		331 / 43 / 46
Records, train / val / test	301,386 / 37,673 / 37,673	301,236 / 37,705 / 37,791
Split artifact	pinned_splits_from_010_seed42.json.gz	query_pair_disjoint_splits_025.json.gz
Transformer run	GH 1598	GH 1640, and Table 4

Under arm Q a held-out record’s query pair has never been seen, but its genes have: every array gene of the validation and test parts appears in training, as do 78 of 86 validation and 88 of 92 test query-pair genes, so 81 percent of validation and 92 percent of test records have all three genes in the training set. The split holds out the pair, not the genes.

3 The models ✗

Let record n have perturbed set S_n of three genes, encoded as $x_n \in \{0, 1\}^{|G|}$ over the $|G| = 4,352$ genes, with label $y_n = \tau$. A model is a set function $f : 2^G \rightarrow \mathbb{R}$, and its capacity to represent interaction is a structural property of the hypothesis class: for distinct genes g, h, k and any S disjoint from them,

$$\Delta_{gh}f(S) = f(S \cup \{g, h\}) - f(S \cup \{g\}) - f(S \cup \{h\}) + f(S), \quad \Delta_{ghk}f(S) = \sum_{U \subseteq \{g, h, k\}} (-1)^{3-|U|} f(S \cup U). \quad (2)$$

A model with $\Delta_{ghk}f \equiv 0$ cannot express a three-way interaction and is a null with respect to the label, whatever it is fit to.

B0, train mean. $\hat{y} = \bar{y}_{\text{train}}$. The zero point.

B1, additive per-gene ridge. $\hat{y}(S) = \beta_0 + \sum_{g \in S} \beta_g$, with β the ridge minimizer $(X^\top X + \alpha I)^{-1} X^\top \hat{y}$ on the centered training labels and α chosen on validation over $\{10^{-2}, \dots, 10^3\}$. Both finite differences vanish identically. The direct analog of the additive null of Visani, Verma and DeWitt (bioRxiv 2026.04.23.719915).

B2, additive plus recurring pairs. B1 with one coefficient γ_{gh} per query pair in the training split. Each γ_{gh} is the mean additive residual over that screen, shrunk by $\alpha/(n_{gh} + \alpha)$, so B2 is B1 plus one nearly unshrunk offset per screen. Second-order capacity on observed pairs, no third-order capacity.

B3, screen mean. No fitted parameters. The mean training label over the records carrying the record’s query pair; if no pair in the record recurs in training, the mean of its three gene means; failing that, the train mean. On arm R the first case always fires; on arm Q it never does.

B4, query pair only. B2 without the gene terms: a per-screen offset and nothing else. The array gene contributes nothing, so B4’s score is an upper bound on how much of the metric is the identity of the screen. On arm Q no test record carries a pair with a coefficient, so B4 is structurally zero.

B5, nonlinearity on the same features. A free 128-vector per gene, summed over the three perturbed genes and read out by a two-hidden-layer network, trained by gradient descent with early stopping on validation, three seeds. Exactly B1’s feature space, but $\Delta_{ghk} \neq 0$, so B5 is not a null; it is the smallest departure from B1 that has interaction capacity.

The transformer. The cell graph transformer of experiment 010, run on the 025 build without change in architecture, loss, optimizer or schedule (cgt_s0_r_k1_000 for arm R, cgt_s0_q_k1_004 for arm Q). Its strain-dependent computation is a set function over the three perturbed gene identities, the same input the nulls see; the nine interaction graphs reach the prediction only through the trained weights, via a penalty on layer-1 attention. The long reading of that code is in the 010 document and is not repeated.

Table 2: What each model can represent. The last two columns are properties of the hypothesis class, not fitted results. Only B5 and the transformer can express a three-way interaction, which is what the label is.

Model	Per-gene quantity	Set aggregation	Fit by	$\Delta_{gh} \neq 0$	$\Delta_{ghk} \neq 0$
B0	none	none	train mean	no	no
B1	free scalar	sum	ridge	no	no
B2	free scalar	sum, plus one offset per recurring pair	ridge	yes	no
B3	label mean over training triples	mean over pairs, else genes	label means	yes	no
B4	none	one offset per recurring pair	ridge	yes	no
B5	free 128-vector	sum, then MLP	gradient descent	yes	yes
CGT	free 180-vector through an encoder	attention over S , mean, MLP	gradient descent	yes	yes

Comparability. Every model is fit on the training part of the arm it is reported under, selects its free choices on the validation part, and is reported on test. The transformer’s label normalizer is fit on the training part for arm Q and on all of S0 for arm R; Pearson is invariant to that choice, and the nulls use the training mean only.

Table 3: Test Pearson for every model under both arms, beside the 010 numbers. B0 through B4 are closed-form fits on a fixed split and carry no spread; B5 is the mean and standard deviation over three seeds. The three transformer rows are on different footings and the row label says which: the 010 checkpoints were scored on the 010 test split; GH 1598 and GH 1640 are validation maxima over 36 logged epochs, and neither has a test score.

Model	Arm R, random over records		Arm Q, query-pair disjoint
	010 build	025 build	025 build
B0 train mean	0.000	0.000	0.000
B4 query pair only	0.390	0.390	0.000
B3 screen mean	0.390	0.390	0.142
B1 additive per-gene ridge	0.400	0.400	0.185
B2 additive plus pair ridge	0.406	0.406	0.180
B5 embedding MLP, 3 seeds	0.426 ± 0.001	0.432 ± 0.005	0.141 ± 0.007
CGT, three 010 checkpoints, test	0.438 to 0.455		
CGT GH 1598, val max, epoch 14 of 36		0.446	
CGT GH 1640, val max, epoch 7 of 36			0.199

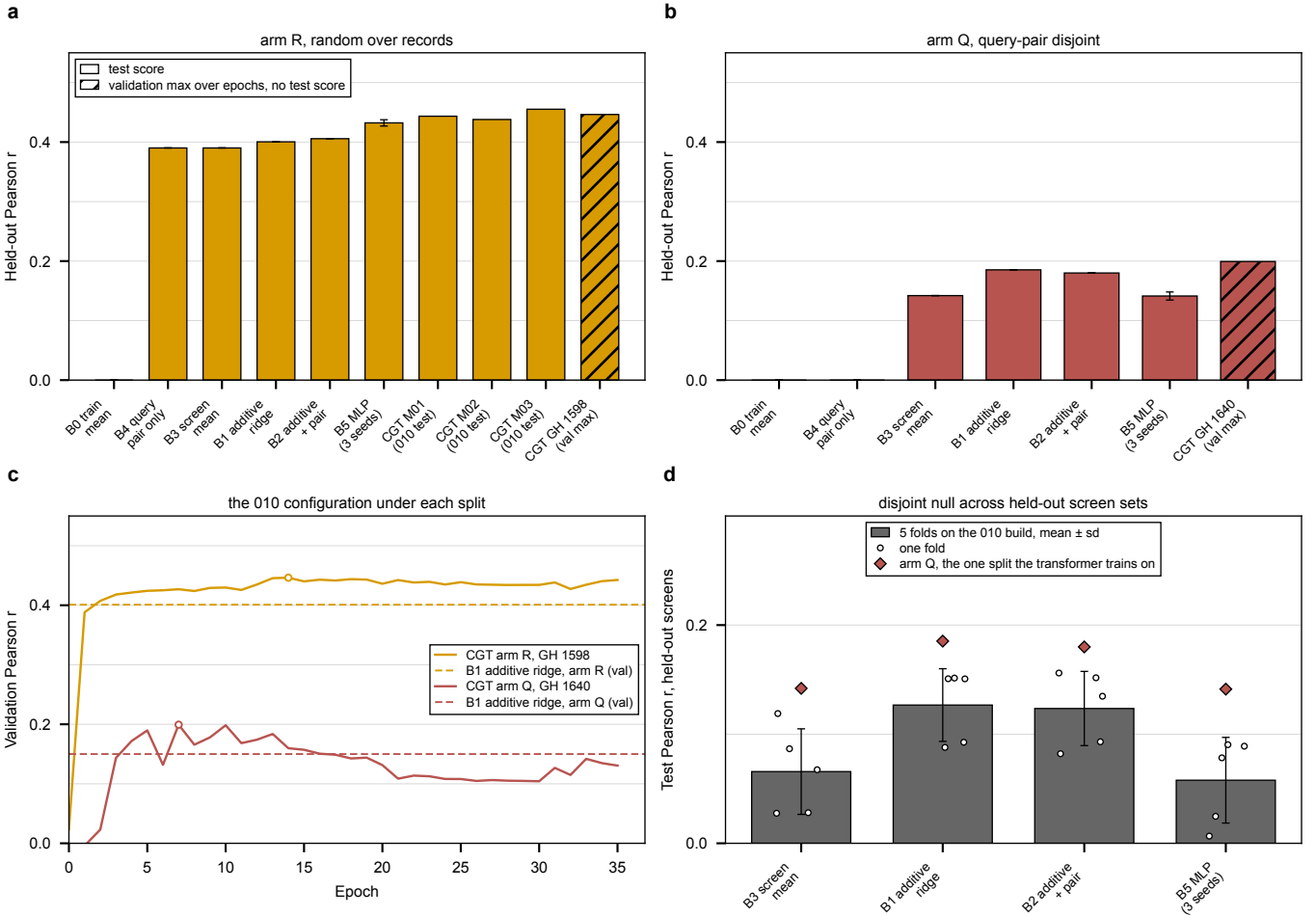


Figure 1: The ladder under both splits. **a**, arm R, random over records: the six baselines on test, the three 010 checkpoints on the 010 test split, and the 025 replication GH 1598 as a validation maximum, hatched. **b**, arm Q, query-pair disjoint, same layout, with GH 1640 hatched. Only B5 carries an error bar, the standard deviation over its three seeds; every other bar is one deterministic fit. **c**, validation Pearson per epoch for the two arms of the 010 configuration, each against the additive ridge fit on its own validation part; open circles mark each run's best epoch, the value the hatched bars carry. **d**, how much the disjoint null moves with the choice of held-out screens: five disjoint folds on the 010 build per model, mean and standard deviation with each fold as a point, and the arm Q single split as a diamond. B0 and B4 are zero on every fold and are omitted.

4 The ladder under both splits ✗

Arm R is the 010 result. Every closed-form model agrees with its 010 value to 3×10^{-13} , which follows from the record-level label and split identity of Section 2 rather than testing it. B5 differs by 0.006 against a three-seed standard deviation of 0.005; the 025 fit ran on CPU where 010's ran on GPU, and the cause is untraced. GH 1598 reaches a validation maximum of 0.446 at epoch 14, inside the 0.447 to 0.462 band of the three 010 checkpoints' validation scores.

On the random split the ladder is mostly screen identity. B4 knows only which query pair a triple carries and reaches 0.390, 97 percent of the additive ridge and 86 percent of the best checkpoint. The transformer's margin over B1 is real, +0.04 to +0.055 with bootstrap intervals excluding zero in the 010 analysis, and small.

Holding out screens collapses the ladder and inverts it. On arm Q the additive ridge falls from 0.400 to 0.185, B4 to zero by construction, and B5 from 0.432 to 0.141, below the ridge. Extra capacity that helped on unseen records of seen screens hurts on unseen screens. B3 falls back from a screen mean to a gene mean and lands at 0.142.

One disjoint split is a small sample of screens. Panel d is why arm Q's numbers are read as one draw rather than as the disjoint null. Across five disjoint folds on the 010 build the ridge spans 0.088 to 0.151, 0.127 ± 0.033 , and arm Q's 0.185 lies above every fold. The same holds for B2, B3 and B5, so arm Q is a comparatively easy draw of 46 held-out screens. Any transformer number on arm Q is compared to the arm Q nulls, on the same 46 screens, and not to the fold mean.

5 The transformer on held-out screens ✗

Four transformer runs have trained on arm Q. One is the 010 configuration. The other three come from the joint-fitness-head branch (PR #346, not yet on main): they replace the learnable gene table with a fixed sequence embedding of matched parameter count, read out through a perturbed CLS token, and train at a constant learning rate for 30 epochs. They share the split and the graph penalty with GH 1640 and nothing else that matters for a replicate, so they are four runs on one split, not one run replicated.

Table 4: Every transformer run on arm Q. Epochs is the number logged; GH 1640 was stopped by the 12 hour wall clock partway through epoch 36, the IGB runs finished their 30. The validation maximum is an upward-biased order statistic over the logged epochs. No run has a test score. Configs of the IGB runs are on branch `feat/025-fitness-joint-head`.

Run	Config	Gene input	Schedule, readout	Epochs	Val max (epoch)	Val last
GH 1640	<code>cgt_s0_q_kl_004</code>	learnable table	cosine, 010 schedule, mean	36	0.199 (7)	0.131
IGB 2391132	<code>cgt_s0_q_kl_emb_017</code>	four-region sequence composite	constant 2.5×10^{-4} , perturbed CLS	30	0.262 (2)	0.222
IGB 2391133	<code>cgt_s0_q_kl_calmb_020</code>	CaLM codon embedding	constant 2.5×10^{-4} , perturbed CLS	30	0.252 (7)	0.208
IGB 2391134	<code>cgt_s0_q_kl_prot_021</code>	ProtT5 protein embedding	constant 2.5×10^{-4} , perturbed CLS	30	0.270 (3)	0.135

Every run peaks early and falls. GH 1640 passes the additive validation null of 0.150 in epoch 3, peaks at 0.199 in epoch 7, falls back through the null near epoch 17 and ends at 0.131, below every null but B3. The three IGB runs peak higher, 0.252 to 0.270 in epochs 2 to 7, and decline from there: the composite and CaLM embeddings end at 0.222 and 0.208, still above the ridge, and ProtT5 ends at 0.135, level with GH 1640. Arm R shows the opposite shape, a margin over its null that appears in epoch 2 and holds for the rest of the run (Figure 1c).

What the early peak does and does not say. On validation, then, a transformer holds a margin over additivity for a few epochs on screens it has never seen, and the size of that margin depends on the gene input and the schedule. Two things keep this from being a result. The maxima are order statistics over 30 to 36 noisy epochs, so each is biased upward by an amount the nulls, which are single fits, do not share. And validation is 43 screens; the test part is a different 46, and Figure 1d shows how far one draw of screens moves the null. A checkpoint scored on test at its validation-selected epoch is the number that compares to Table 3, and no run has one.

Hypothesis, untested. The decline after the peak is consistent with the model fitting screen-specific structure of the training pairs that does not transfer, the same structure B4 measures, but no measurement here separates that from ordinary overfitting of the label noise, and the mechanism is not claimed.

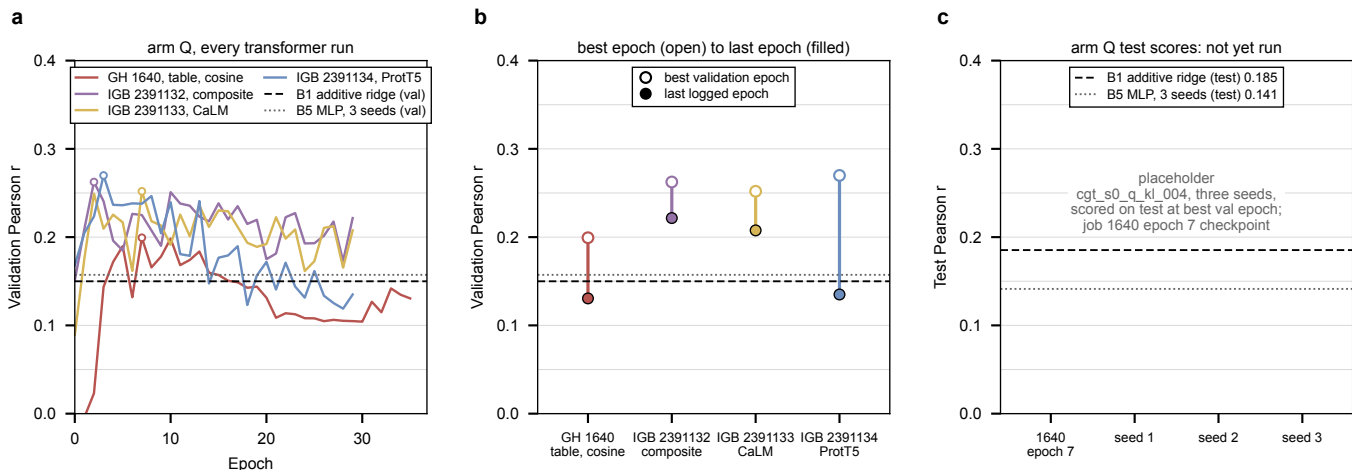


Figure 2: Every transformer run on the query-pair-disjoint split. **a**, validation Pearson per epoch, with the additive ridge and the three-seed MLP fit on the same validation part as horizontal lines; open circles mark each run’s best epoch. **b**, the same runs reduced to their best epoch (open) and last logged epoch (filled). **c**, placeholder for the test comparison that has not been run: the arm Q test nulls are drawn, and the empty slots are the epoch 7 checkpoint of GH 1640 and three seeds of the 010 configuration, each to be scored on test at its best validation epoch.

6 What is left before this enters the main text ✗

- Score GH 1640’s epoch 7 checkpoint on the arm Q test part.** The checkpoint survives (`$DATA_ROOT/models/checkpoints/327csnlk-best-pearson-epoch=07-val*`). One scoring pass fills the first slot of Figure 2c and gives the first transformer test number against B1 0.185 and B5 0.141.
- Replicate the 010 configuration on arm Q.** Three seeds of `cgt_s0_q_kl_004` with a test evaluation at the best validation epoch, under one fixed epoch budget. GH 1640 shows the run needs about 10 epochs to reach its peak and 36 epochs cost the 12 hour wall clock, so a 20 epoch budget with checkpoint-on-best is enough. These fill the remaining slots of Figure 2c and give the transformer row of Table 3 a spread.
- Decide whether the constant-learning-rate variants belong in the ladder.** The IGB runs are a different model, readout and schedule. If they are to be compared to the nulls they need the same treatment as item 2: a learnable-table control at constant learning rate on arm Q (`cgt_s0_q_kl_ctr1_016`, which has no run on W&B as of 2026-09-11), seeds, and test scores. Otherwise they stay in Figure 2a,b as validation trajectories and are not in Figure 1.
- A paired bootstrap on arm Q test.** The 010 analysis established the random-split margin with a paired bootstrap over test records. The same 2,000-resample interval on the arm Q predictions is what turns a transformer minus B1 difference into a claim.
- Fold spread for the transformer, or an explicit statement that it is unaffordable.** Panel d of Figure 1 is five folds for the nulls. A transformer fold costs about 12 hours of one GilaHyper node per 36 epochs, so five folds by three seeds is 15 runs. If the main text reports arm Q alone it should say so and cite the fold spread of the null as the scale of the split-to-split variation.
- B5 on GPU.** The 025 B5 fit ran on CPU and differs from 010’s GPU fit by 0.006 on arm R. A GPU refit closes the one row of Table 3 that is not an exact reproduction.
- Bibliography.** The Visani preprint and Kuzmin 2018 are named by DOI in prose. A Zotero collection for this document, then `make bib-pull`, makes them citations.

What this document does not carry over from the 010 report. The reading of the transformer code that establishes it is a set function over the perturbed gene identities, the graph penalty’s share of the loss, the neighbor recall of the regularized heads, the matched pair of runs without the penalty, the record-by-record agreement between models, and the gene-index defect of the inference runs. Each is in `notes-tex/010-additive-baselines` and is cited from there when the main text needs it.